

Homework 1

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Problem 1

$$(a) 2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$$

[ANSWER: 24]

$$(b) (1/2)w = [-2, 1, 3]^T$$

$$v + (1/2)w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$\|v + (1/2)w\| = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3} \approx 5.20$$

[ANSWER: $3\sqrt{3} \approx 5.20$]

$$(c) \cos \theta = (u \cdot w) / (\|u\| \|w\|)$$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$\|u\| = \sqrt{4 + 1 + 9} = \sqrt{14}, \quad \|w\| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

~~$\cos \theta = 8/\sqrt{56}$~~ no, the denominator is both norms multiplied:

$$\cos \theta = 8 / (\sqrt{14} \cdot 2\sqrt{14}) = 8/28 = 2/7$$

$$\theta = \arccos(2/7) \approx 73.40^\circ$$

[ANSWER: $\theta \approx 73.40^\circ$]

Problem 2

(a) AB is not defined. A is 3×3 and B is 2×3 , and A 's 3 columns do not match B 's 2 rows.

BA is $(2 \times 3)(3 \times 3) = 2 \times 3$:

row 1: $[1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$

row 2: $[3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$

$$BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 4 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 4 & -1 \end{bmatrix}$$

(b) $Av = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$

vA is defined too. Multiplication is commutative for this, you are hitting the same matrix with the same vector, so vA is just Av laid on its side:

$$vA = Av^T = \begin{bmatrix} 4 & -11 & -5 \end{bmatrix}$$

(c) Aw : not defined.

WA is $(1 \times 3)(3 \times 3) = 1 \times 3$:

$WA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)] = [4, -6, -1]$

(d) vw is $(3 \times 1)(1 \times 3) = 3 \times 3$:

$$vw = \begin{bmatrix} 2 & 0 & -1 \\ -4 & 0 & 2 \\ 6 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 0 & -3 \end{bmatrix}$$

wv is $(1 \times 3)(3 \times 1) = 1 \times 1$: $2(1) + 0(-2) + (-1)(3) = -1$, so $wv = [-1]$.

(e) $A^2 = AA$:

row 1: $[2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$

row 2: $[1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$

row 3: $[0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 16, -7]$

$$A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$$

(f) B^2 is not defined: B is 2×3 , and squaring needs a square matrix (3 columns would have to meet 2 rows).

Problem 3

(a) $\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 2 & 3 & 1 & | & 2 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & -4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 10 & | & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$x_3 = 1, x_2 = -4 + 3(1) = -1, x_1 = -1 - 2(-1) + 1 = 2$, and there are no free columns

so the solution is unique.

ANSWER: $x = [2, -1, 1]^T$

(b) $\begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$

$$[2 \ 5 \ 1 \mid 6]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

$$R_1 \rightarrow R_1 - 2R_2: [1 \ 0 \ -7 \mid 8]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

Column 3 has no pivot, so z is the free variable. Let $z = t$, then $x = 8 + 7t$ and $y = -2 - 3t$. The solution set is a line, so there are infinitely many solutions.

$$[\text{ANSWER: } [x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}]$$

Problem 4

(a) Let x_1 = kg of Alloy 1 used, x_2 = kg of Alloy 2 used, x_3 = kg of Alloy 3 used (masses

in kilograms). The 100 kg of product is 36 kg copper, 44 kg zinc, 20 kg nickel, and no

metal is lost, so each metal balances:

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

(b) Multiply through by 10 so there are no decimals:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$(c) [5 \ 2 \ 4 \mid 360] \quad [1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440] \rightarrow [3 \ 7 \ 2 \mid 440] \rightarrow [0 \ 7 \ 14 \mid 560]$$

$$[2 \ 1 \ 4 \mid 200] \quad [2 \ 1 \ 4 \mid 200] \quad [2 \ 1 \ 4 \mid 200]$$

$$[1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40] \\ \rightarrow [0 \ 7 \ 14 \mid 560] \rightarrow [0 \ 1 \ 2 \mid 80] \rightarrow [0 \ 1 \ 2 \mid 80]$$

$$[0 \ 1 \ 12 \mid 280] \quad [0 \ 1 \ 12 \mid 280] \quad [0 \ 0 \ 10 \mid 200]$$

$$[1 \ 0 \ -4 \mid -40] \\ \rightarrow [0 \ 1 \ 2 \mid 80] \\ [0 \ 0 \ 1 \mid 20]$$

$x_3 = 20$, $x_2 = 80 - 2(20) = 40$, $x_1 = -40 + 4(20) = 40$. Total is 100 kg, which checks.

[ANSWER: 40 kg of Alloy 1, 40 kg of Alloy 2, 20 kg of Alloy 3]

Problem 5

(a) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 3 & 6 & 4 \end{bmatrix} \quad R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 3 & 6 & 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 3 & 6 & 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_3: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_1 \rightarrow R_1 - R_3: \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2: \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

[ANSWER: $\text{RREF}(A) = I$]

(b) $\text{rank}(A) = 3$ because the RREF is the identity, so there is a pivot in all three columns.

[ANSWER: $\text{rank}(A) = 3$]

(c) ~~it depends on b, since~~ Actually it does not depend on b at all here.

A row reduces all the way to 1, so $[A \mid b]$ row reduces to $[I \mid c]$ for some column c, and reading that off gives x_1, x_2, x_3 directly with nothing left over. So $Ax = b$ has

exactly one solution, no matter which b you pick.

0 solutions: impossible. Every row of the RREF has a pivot in it, so you can never end

up with a row that reads $0 = \text{nonzero}$, which is the only way a system is inconsistent.

Infinitely many: impossible. All three columns are pivot columns, so there is no free

variable, and without a free variable you cannot build a second solution.

[ANSWER: always exactly 1 solution; 0 and infinitely many are both impossible]

Problem 6

(a) Row reduce the matrix whose columns are v_1, v_2, v_3 :

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_2: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2: \begin{bmatrix} 1 & 0 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Two pivots but three vectors, so they are dependent. The relation is $v_3 = v_2 - 2v_1$, since $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$. Two of them are independent, so the span

is 2-dimensional and passes through the origin.

[ANSWER: dependent; span is a plane]

(b) $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$

$R_2 \rightarrow R_2 - R_1: \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2: \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & -1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 & 2 \end{bmatrix}$

Three pivots for three vectors, so $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ only when all the c 's are

0. That is independence, and three independent vectors fill up $\mathbb{R}^3$.

[ANSWER: independent; span is all of $\mathbb{R}^3$]